LESSON PLAN

BASIC ELECTRICAL (23)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	10
Total No. of Lesson Plans	29	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Alighting gear, basic design requirements, types of undercarriage and functions
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Alighting gear, basic design requirements, types of undercarriage and functions
5. Specific Objectives	23.1.3.1 Trainee will be able to describe typical alighting gears of an aircraft and its construction.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Examination of undercarriage models; Safety: Handle models with care to prevent damage; Supervision: Instructor monitors handling of components
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of the alighting gear?
(b) Name the three main types of undercarriage configurations.
(c) Why is shock absorption necessary during landing?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Basic design requirements for alighting gear • Load distribution and weight bearing • Shock absorption mechanisms • Braking systems integration • Steering and turning mechanisms	L & Ds	PPT, IB	15 min
(b)	• Types of undercarriage: Conventional vs Tail-dragger • Tricycle configuration analysis • Retraction mechanisms overview • Wheel types and tire construction • Brake actuation systems	L & Ds	PPT, PDF	20 min
(c)	• Functions of alighting gear in flight operations • Ground handling limitations • Emergency extension procedures • Maintenance considerations • Safety protocols for gear operation	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the difference between conventional and tricycle undercarriage configurations.
(b) Describe the role of shock absorbers in the alighting gear system.
(c) List two critical safety checks performed before landing.

12. Home Assignment and Follow Up	Prepare a brief note on the typical circuit diagram for undercarriage retraction and extension to be discussed in LP 11.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Safety and emergency devices for retraction and extension, typical u/c circuits,	Refer to the master syllabus reference list	LP 11

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min